Person 3 Presentation Notes DCRNN Evidence and Conclusion

Purpose: study these notes before presenting the assigned slides. Explain the ideas in your own words and use the examples when the audience is unfamiliar with AI or power grids.

What this person must understand

This section explains the advanced research models, the original figures in the results folder, why DCRNN did not win this run, how the frontend and backend connect, and what the group can claim honestly.

DCRNN in one paragraph

DCRNN means Diffusion Convolutional Recurrent Neural Network. It reads a sequence of previous hourly observations. Before the recurrent memory update, it passes information through the graph so a country can receive context from connected countries. It then predicts the next net flow for all graph nodes together. A road analogy is useful: before predicting traffic on one road, the model also checks nearby roads.

Slide-by-slide explanation

Slide 9 DCRNN in simple terms

Purpose: Explain graph diffusion and compare the saved result.

- Fusion GRU reads a sequence with local and global features.
- DCRNN preserves the node dimension and passes signals through forward and reverse graph transitions.
- DCRNN is a benchmark implementation, not a dispatch solver.

What to say: Graph diffusion lets neighboring countries share information before the recurrent update. It tests whether network context adds predictive value beyond local history.

Slide 10 Saved results and interpretation

Purpose: Explain why each original figure matters.

- Training curves show whether learning is stable and whether validation behavior follows training behavior.
- ROC and precision-recall curves show performance across classification thresholds.
- Feature importance shows how much validation error rises when a feature is shuffled.
- Error analysis shows both stable cases and abrupt failures.

What to say: The figures are saved experiment outputs. Recent local net flow is the strongest signal in this run. Stable flow is easier to predict; abrupt changes and class-boundary changes are harder.

Slide 11 Saved results: feature interpretation and error analysis

Purpose: Explain what the model learned and where it fails.

- Feature importance shows how much validation MAE increases when one feature is shuffled.
- Recent local net flow is the strongest saved signal; network centrality and residual features contribute less in this run.

- The error distribution shows many smaller errors and a long tail of difficult cases.
- These results explain model behavior; they do not prove causal relationships.

What to say: The model is mainly using recent local flow to make its next-hour estimate. Sudden changes and unusual operating conditions create the larger errors.

Slide 12 Complete workflow, novelty, and limits

Purpose: Close the presentation with one honest end-to-end explanation.

- CSV files feed the Python preprocessing and feature-engineering pipeline.
- The pipeline saves processed artifacts, model results, and figures for reproducibility.
- Backend services provide snapshots, forecasts, metrics, loss estimates, and route plans.
- The frontend presents the graph, replay animation, charts, and what-if routing result.
- The novelty is the auditable integration of data quality, graph features, model comparison, proxy physics, and route guardrails.
- The project still needs verified ratings, N-1 security analysis, calibrated uncertainty, and a real telemetry adapter before operational use.

What to say: Prediction estimates what may happen. Routing suggests what should be investigated. Engineering validation decides whether any action is safe. We should not claim a brand-new neural architecture.

How to explain the original figures

Figure	Meaning	Safe conclusion
model_comparison.png	RMSE and F1 across baselines and trained models	Fusion GRU has the strongest saved trained RMSE; Persistence remains the operational selection artifact.
training_curves.png	Fusion GRU learning and validation behavior	Check the training-validation relationship, not only the final training value.
dcrnn_training_curves.png	DCRNN loss and macro-F1 over epochs	Graph training is measurable, but graph context did not win this run.
roc_pr_curves.png	Threshold behavior for stress classes	Useful for understanding false alarms and missed stress.
feature_importance.png	Validation MAE increase after shuffling features	Recent local flow dominates the saved model's predictive signal.
tsne_projection.png	Two-dimensional view of learned hidden states	Similar learned states may group, but this does not prove causality.
error_distribution.png	Distribution of prediction errors	Large or asymmetric errors identify conditions needing more data or a stronger model.

Questions you should be ready for

- Why include DCRNN if it did not win? A fair negative or weaker result is still evidence about this dataset and protocol.
- What is novel? Transparent integration and evidence-driven model selection, not an unsupported new architecture.
- Can DCRNN control the grid? No. It forecasts flow; it does not authorize switching or dispatch.

- What would make this deployable? Verified physical parameters, live telemetry, N-1 validation, calibrated uncertainty, and operator approval.

Final answer to remember

GridPulse turns raw hourly records into a graph, creates a 24-hour model window, predicts next-hour net flow, classifies stress, estimates proxy loss, and tests whether an alternate path could reduce pressure. It is a transparent decision-support prototype with clear engineering limitations.